

2D Laplace equation solved with the finite element method

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1 the equation

We solve the 2D Poisson equation

$$-\nabla^2 \Psi = \delta(\mathbf{r} - \mathbf{r}_c), \quad (1)$$

with the point charge localized in point $\mathbf{r}_c$. We will assume that $\mathbf{r}_c$ is outside the computational box $[0, L] \times [0, L]$. The (1) equation is solved by 2D Coulomb potential is

$$\Psi(\mathbf{r} - \mathbf{r}_c) = -\frac{1}{2\pi} \ln(|\mathbf{r} - \mathbf{r}_c|) \quad (2)$$

We want to use finite element method to tackle the problem numerically. We use a basis of shape functions

$$\Psi(x, y) = \sum_{i=1}^N c_i h_i(x, y), \quad (3)$$

where h_i is associated with a node i , e.g. for a node coordinate $\mathbf{p}_j$ we have $h_i(\mathbf{p}_j) = \delta_{ij}$.

The vector of coefficients c solves a linear system of equations $\mathbf{S}c = \mathbf{b}$ with the stiffness matrix $\mathbf{S}$ and the right-hand-side vector $\mathbf{b}$ that in our case is determined by the boundary conditions.

2 the elements and nodes

We generate N^2 identical elements in a computational box of size $L \times L$. Let $L = 5$. Each element is a square with side of length $a = L/(N)$. The elements are indexed by an integer from 1 to $(N)^2$. We use first bilinear shape functions (replacing g_i instead h_i in Eq. (3)) and next biparabolic ones. We have $(N + 1)^2$ nodes (in red in Fig. 1) for bilinear shape functions and $(2N + 1)^2$ nodes for a biparabolic shape functions (both red and black in Fig. 1). The nodes needs to be numbered and their (x, y) coordinates stored in a matrix $p(l, o)$, where $l = 1$ is the node index and $o = 1, 2$ for x and y coordinates, respectively. Moreover in each element we need to number the

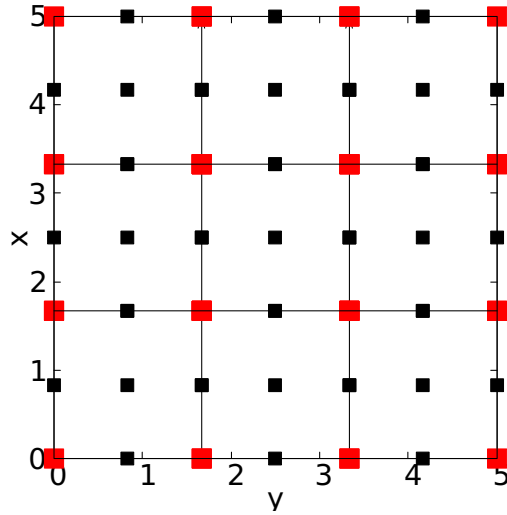


Figure 1: The elements and nodes for $N = 3$, and $L = 5$, the corner nodes are marked in red, the other ones in black. For the bilinear functions we use the red nodes and for the biparabolic nodes both black and red ones.

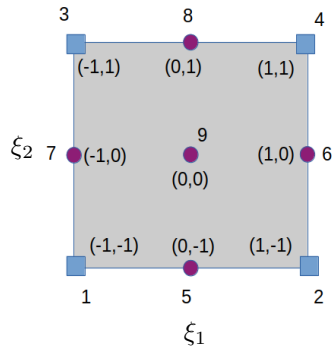


Figure 2: Each element in its reference space is a square of side length 2. The local indexing of the nodes and their coordinates (ξ_1, ξ_2) . For bilinear basis we use shape functions for nodes from 1 to 4, and for biparabolic functions the nodes from 1 do 9.

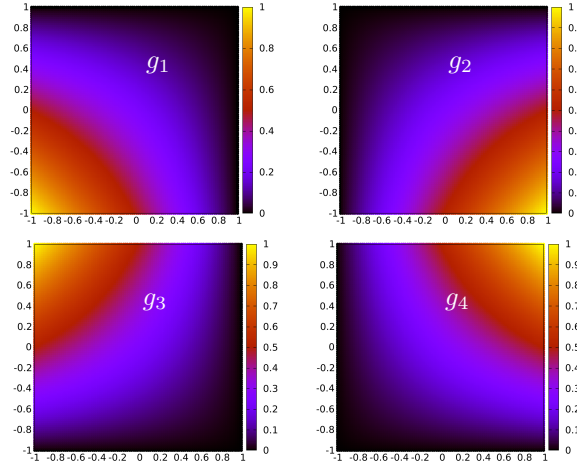


Figure 3: Bilinear shape functions. $g_k(\vec{\xi}_i) = \delta_{ki}$

nodes locally from 1 to 4 for bilinear and from 1 to 9 for biparabolic shape functions respectively (Fig. 2).

We will also need a table that stores the global node number for a node of a local number i of the element k that we will denote as $nlg(k, i)$.

The mesh of nodes and the indexing can be produced in a number of ways. The lecturer uses an approach similar to generation of a crystal lattice. Let us imagine that each of the element is a crystal cell. The crystal structure is generated by nodes of local number 1 (left, bottom), and the rest of the nodes is generated by 3 or 8 basis vectors of our "crystal" structure for bilinear and biparabolic functions. So, in the context of Fig. 1 we generate the red nodes for x and $y \in [0, 3\frac{1}{3}]$. Generation of each of this nodes generates the element. Then for each of the element we generate 3 (or 8) extra nodes in positions shifted with respect to the node of local index by a basis vector:

$$\begin{aligned} b_2 &= (a, 0) & b_3 &= (0, a) & b_4 &= (a, a) \\ b_5 &= (a/2, 0) & b_6 &= (a, a/2) & b_7 &= (0, a/2) & b_8 &= (a/2, a), b_9 = (a/2, a/2) \end{aligned}$$

We check whether a node already exists in the position of a local node we want to generate (k local index in the element m): since it could already have been generated. Upon the check we either produce the next node of a global number n or find the index of the node that already exists at the required position. Lets denote the index by n in the matrix $\mathbf{p}_n$. In either case we store the information that $nlg(k, m) = n$.

3 elements and shape functions in the reference space

In FEM one uses a reference space for evaluation of the matrix element. In the reference space each element is a square of coordinates $\vec{\xi} = (\xi_1, \xi_2) \in [-1, 1] \times [-1, 1]$.

For element k the transfer from the reference space to the real space has the form

$$x(\xi_1, \xi_2) = \sum_{i=1}^4 x_{nlg(k,i)} g_i(\xi_1, \xi_2), \quad (4)$$

$$y(\xi_1, \xi_2) = \sum_{i=1}^4 y_{nlg(k,i)} g_i(\xi_1, \xi_2), \quad (5)$$

where $(x_l, y_l) = \mathbf{p}_l$ are the real space coordinates of the corner node of a global index $l = nlg(k, i)$, where $i = 1, 2, 3$ or 4 , and g_i are the bilinear shape functions defined as

$$g_1(\vec{\xi}) = f_1(\xi_1) f_1(\xi_2) \quad (6)$$

$$g_2(\vec{\xi}) = f_2(\xi_1) f_1(\xi_2) \quad (7)$$

$$g_3(\vec{\xi}) = f_1(\xi_1) f_2(\xi_2) \quad (8)$$

$$g_4(\vec{\xi}) = f_2(\xi_1) f_2(\xi_2), \quad (9)$$

with

$$f_1(\xi) = \frac{1 - \xi}{2} \quad (10)$$

$$f_2(\xi) = \frac{1 + \xi}{2}. \quad (11)$$

We have $g_i(\vec{\xi}_l) = \delta(i, l)$ (Fig. 3).

The bilinear functions g_i are always used for the mapping from the reference to the physical space. The solution of the Laplace equation can be found for the bilinear functions or for the biparabolic functions (Fig. 4). The latter are defined as

$$h_1(\vec{\xi}) = q_1(\xi_1) q_1(\xi_2) \quad (12)$$

$$h_2(\vec{\xi}) = q_3(\xi_1) q_1(\xi_2) \quad (13)$$

$$h_3(\vec{\xi}) = q_1(\xi_1) q_3(\xi_2) \quad (14)$$

$$h_4(\vec{\xi}) = q_3(\xi_1) q_3(\xi_2) \quad (15)$$

$$h_5(\vec{\xi}) = q_2(\xi_1) q_1(\xi_2) \quad (16)$$

$$h_6(\vec{\xi}) = q_3(\xi_1) q_2(\xi_2) \quad (17)$$

$$h_7(\vec{\xi}) = q_1(\xi_1) q_2(\xi_2) \quad (18)$$

$$h_8(\vec{\xi}) = q_2(\xi_1) q_3(\xi_2) \quad (19)$$

$$h_9(\vec{\xi}) = q_2(\xi_1) q_2(\xi_2), \quad (20)$$

z

$$q_1(\xi) = \xi(\xi - 1)/2 \quad (21)$$

$$q_2(\xi) = (1 - \xi)(1 + \xi) \quad (22)$$

$$q_3(\xi) = \xi(\xi + 1)/2. \quad (23)$$

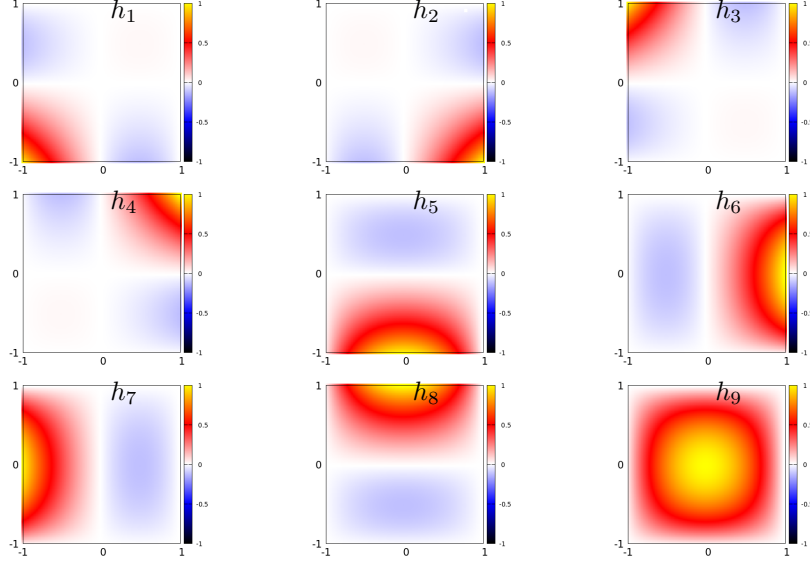


Figure 4: Biparabolic shape functions.

In the element k our solution to the Laplace equation is spanned on 4 shape functions g or 9 shape functions h . In the latter case the solution in point $\vec{\xi} \rightarrow \vec{r}$ in element k is given by

$$\Psi\left(\vec{r}(\vec{\xi}) \in \Omega_k\right) = \sum_{i=1}^{4/9} \Psi_{nlg(k,i)} h_i(\xi_1, \xi_2). \quad (24)$$

4 a local stiffness matrix

We need to calculate the local stiffness matrix of size 4×4 or 9×9 depending on the choice of the basis.

$$\mathcal{T}_{ji} = \langle h_j | -\nabla^2 | h_i \rangle = \langle \nabla h_j | \nabla h_i \rangle.$$

We know that $\frac{d}{dx} = \frac{d\xi_1}{dx} \frac{d}{d\xi_1} = \frac{2}{a} \frac{d}{d\xi_1}$ and $dx = \frac{2}{a} d\xi_1$ so

$$s_{ji}^k = \int_{\Omega_k} d\xi_1 d\xi_2 \left(\frac{dh_j}{d\xi_1} \frac{dh_i}{d\xi_1} + \frac{dh_j}{d\xi_2} \frac{dh_i}{d\xi_2} \right).$$

We will use the Gauss quadrature for integration of the stiffness matrix elements

$$s_{ji}^k = s_{ji} = \sum_{l=1}^4 \sum_{n=1}^4 w_l w_n \left(\frac{dh_j}{d\xi_1} \Big|_{\gamma_l, \gamma_n} \frac{dh_i}{d\xi_1} \Big|_{\gamma_l, \gamma_n} + \frac{dh_j}{d\xi_2} \Big|_{\gamma_l, \gamma_n} \frac{dh_i}{d\xi_2} \Big|_{\gamma_l, \gamma_n} \right)$$

with $w_1 = w_2 = (18 + \sqrt{30})/36$, $w_3 = w_4 = (18 - \sqrt{30})/36$ and $\gamma_1 = -\sqrt{\frac{3}{7} - \frac{2}{7}\sqrt{6/5}}$, $\gamma_2 = \sqrt{\frac{3}{7} - \frac{2}{7}\sqrt{6/5}}$, $\gamma_3 = \sqrt{\frac{3}{7} + \frac{2}{7}\sqrt{6/5}}$ i $\gamma_4 = -\sqrt{\frac{3}{7} + \frac{2}{7}\sqrt{6/5}}$.

For our choice of elements s_{ji}^k is the same for them all.

A symmetric finite difference formula differentiates our functions without error.

$$\frac{h_j(\xi_1, \xi_2)}{d\xi_2} \Big|_{\gamma_l, \gamma_n} = \frac{h_j(\gamma_l, \gamma_n + \Delta) - h_j(\gamma_l, \gamma_n - \Delta)}{2\Delta},$$

$\Delta = 0.1$ is a reasonable choice.

Check your local stiffness matrix with the lecturer.

5 Global matrix assembly

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a loop over all elements k
a loop over the local node numbers i1 from 1 to 4 (or 9)
a loop over the local node numbers i2 from 1 to 4 (or 9)
S(nlg(k, i1), nlg(k, i2)) += s(i1, i2)
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6 Boundary conditions

The RHS vector $\mathbf{b}$ is zero for the points from the inside of the computational box due to the lack of internal charges. For the boundary nodes we set the values of the logarithmic exact solution Ψ (let us set $\mathbf{r}_c = (-1, 0)$). For the boundary nodes of a global index i we need to set $c_i = \Psi(\mathbf{p}_i)$. For that purpose we zero all the elements in i -th row of S and set $S_{ii} = 1$. On the RHS we set $\mathbf{b}_i = \Psi(\mathbf{p}_i)$ and we are done with the b.c.

7 deliverables

I. Bilinear functions:

For $N = 4$

1. D.I.1 print position of the nodes and the mapping table nlg. (20 pts)
2. D.I.2 print the local stiffness matrix before the boundary conditions are introduced (20 pts)
3. D.I.3 after solution of the LSE: plot the potential in the nodes and compare it to the exact result (20 pts)

D.I.4. Plot the potential and the difference between the exact and FEM results. We go through the elements. In each element we scan the local coordinates from -1 to 1 with a step of 0.2. We calculate the potential using Eq. (24) and the coordinates x, y in the laboratory frame using Eq. (4,5). (10 pts).

D.I.5. Evaluate the maximal value of the difference between the exact result and FEM result. Plot it as a function of N . (10 pts).

II. Biparabolic functions. For $N = 4$.

1. D.II.1: Repeat D.I.4. (10 pts)
2. D.II.2: Repeat D.I.5. (10 pts)